PHONE CHARGING TRACKING SYSTEM

Face verification + unique charging-card number + device collection control

Purpose	Help a phone-charging shop record devices received, find the correct customer by ticket number, verify the customer when appropriate, and track each device until it is returned.
Primary users	Shop owner / attendant; customer at drop-off and collection.
Core rule	A charging number can have only one active registration at a time. It becomes available again only when every device under that registration is collected and the job is closed.

Project specification and implementation guide

Prototype note: this document describes the intended system and a practical build plan. A real deployment needs secure storage, staff access controls, backups, and a tested face-verification setup. A browser demo alone should not be treated as production-grade biometric security.

1. Main screens

Dashboard

• Show counts for active charging jobs, devices currently in the shop, partially collected jobs, and numbers available for reuse.

• Provide quick actions: New intake, Search charging number, Active jobs, and History.

• Make the current status obvious: Active, Partially collected, Fully collected / Closed.

New intake / registration

• Enter the charging-card number, customer name, and optional contact details.

• Capture a customer face image or perform face verification, with clear notice and consent before capture.

• Choose whether the customer is leaving one device or multiple devices.

• For one device, show a single device form. For multiple devices, let the attendant add as many device rows as needed.

• Each device row should record device type, brand/model, colour, identifying notes or serial/IMEI if appropriate, and condition/accessories received.

• Allow saving the registration and printing or showing a receipt containing the number and device list.

Search and clock out / collection

• Search by charging number. If active, show the original registration details, face thumbnail/verification status, all devices, registration time, and collection status.

• If the number is already active and someone tries to register it again, block the new registration and show the existing record. Offer a safe shortcut to open that record and proceed to collection if appropriate.

• Ask: “Collect all devices now?” If Yes, mark all remaining devices collected after confirmation. If No, show each device with its own Collect button.

• Record who marked each device collected and the date/time. Keep a partial-collection job active until all devices are collected.

• Once every device is collected, close the job and make the number available for a new registration.

Add devices after registration

• Allow an attendant to reopen an active job by searching its number and choose “Add device”.

• Append the new device to the existing job; do not overwrite or remove previously recorded devices.

• Record the staff member, time, and reason/note for the addition.

2. Key business rules

Rule	Expected behaviour
Unique active number	Reject a second active registration for the same number; show the first active record.
Number reuse	Release the number only after all devices in the job are collected and the job is closed.
Partial collection	A collected device is individually marked returned; uncollected devices remain active.
Adding devices	New devices can be added to an active job; audit the change.

Rule	Expected behaviour
Search safety	Never silently replace an existing job or let a new job hide the old one.
Face verification	Use as a supporting identity check, not as the only proof of ownership. Provide a manual fallback for failed matches.
Audit history	Keep a history of closed jobs and collection actions even after the number becomes reusable.

3. Suggested data structure

Use separate records for a charging job and its devices. This supports one number with multiple devices and partial returns.

Table / collection	Important fields
ChargingJobs	**id, charging_number, customer_name, contact_optional, face_template_or_reference, face_consent_at, status, created_at, created_by, closed_at**
Devices	**id, job_id, device_type, brand_model, colour, identifier_optional, condition_notes, accessories, status, added_at, added_by, collected_at, collected_by**
StaffUsers	**id, display_name, role, password_hash / identity-provider ID, active**
AuditEvents	**id, job_id, device_id_optional, staff_id, event_type, event_time, notes**

Important: enforce uniqueness for charging_number among active jobs in the database, not only in the browser. Use a database transaction or equivalent to avoid two attendants registering the same number at the same time.

4. Face capture and privacy

• Tell customers why their face is being captured and obtain their consent before recording it.

• Restrict access to face data to authorised staff. Use HTTPS, access controls, secure backups, and encryption where supported.

• Prefer storing a protected face template or secure reference rather than an ordinary public image URL. Never put biometric data in browser localStorage for a production system.

• Set a retention policy for biometric data and delete it when no longer needed, subject to lawful recordkeeping requirements.

• Keep a manual fallback: staff can compare the customer to the intake record and ask for the charging ticket or other proof. Face matching can make mistakes.

• For a production implementation, select a vetted face-verification provider or carefully evaluated library and test lighting, camera quality, false matches, and spoofing resistance.

5. Suggested technology

• Frontend: HTML, CSS, and JavaScript for the first learning prototype; later React or Svelte if desired.

• Backend: Node.js + Express API for registration, search, device collection, staff authentication, and audit events.

• Database: PostgreSQL or Supabase with constraints for active charging numbers and relations between jobs and devices.

• Face processing: camera capture in a secure browser context and a dedicated, evaluated verification approach. Camera access generally requires HTTPS or localhost.

• Deployment: HTTPS hosting, environment variables for secrets, database backups, role-based staff accounts, and monitoring.

6. Example workflow

1. Attendant opens New Intake and enters charging number CH-014.

2. Attendant enters the owner's name, captures the face with consent, chooses "Multiple devices", and adds two devices.

3. System checks the number. If CH-014 already has an active job, it rejects the duplicate and displays the existing owner and device list.

4. Customer returns and gives CH-014. Attendant searches the number and reviews the matching job.

5. System asks whether all devices will be collected. If No, the attendant clicks Collect beside the phone being returned.

6. The job remains active while any device is still in the shop. The attendant may add a device to that active job if needed.

7. After the final device is marked collected, the job closes and CH-014 becomes reusable. The historical record remains available.

7. Build and test checklist

• Register one device under a free number.

• Register several devices under one number.

• Attempt duplicate active registration and confirm it is blocked with the original job displayed.

• Search a number that does not exist and show a clear "No active job found" message.

• Collect all devices at once and verify the number becomes available.

• Collect devices one at a time and verify the job stays active until the last device is returned.

• Add a device to an active job and verify the previous records remain intact.

• Test face capture denied, poor lighting, failed match, and manual fallback.

• Test two attendants attempting the same number concurrently.

• Verify audit history, permissions, backup/restore, and deletion/retention policies.

8. Implementation stages

• Stage 1 — Interface prototype: dashboard, intake form, search, multi-device list, per-device collection, and number reuse using sample data.

• Stage 2 — Backend and database: authentication, persistent jobs/devices, duplicate protection, timestamps, audit history, and backups.

• Stage 3 — Camera and face verification: consent, capture, verification workflow, fallback, and privacy controls.

• Stage 4 — Production readiness: HTTPS, staff roles, security review, data-retention policy, concurrent-use testing, and deployment.

This PDF is the project blueprint. A functional website prototype is a separate deliverable from this document; face recognition and shared multi-device records require backend integration before the system is production-ready.